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Challenges in coverage of future hepatitis C vaccines: review and potential solutions

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Abstract

Background: Motivated by the high mortality burden of hepatitis C virus (HCV) and the unprecedented rapid development of the COVID-19 and respiratory syncytial virus (RSV) vaccines, we note that a prompt HCV vaccine rollout may streamline the World Health Organization's goal to eliminate HCV before 2030. While progress in the development of HCV vaccine candidates has rapidly flourished, vaccine hesitancy and HCV incidence are both particularly prevalent in people who inject drugs (PWID). The aim of this paper is to document several potential challenges in HCV vaccine uptake and provide a set of preliminary recommendations for public and community health professionals to improve acceptance.

Methods: We conducted a forward-looking integrative narrative review and identified relevant articles from PubMed. We survey literature discussing barriers to vaccine acceptance in past rollouts (e.g., COVID-19, hepatitis B) and barriers to HCV management, particularly in PWID.

Results: Six key challenges were identified: (1) structural and social barriers affecting PWID, (2) vaccine safety, efficacy, and relevance concerns, (3) multiple-dose attrition and vaccine fatigue, (4) media presentation and misinformation, (5) awareness and attitude towards infection, and (6) information framing and primary care linkage. Four possible recommendations were also

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Author Contributions

Conceptualization, K.G., K.P., H.D., and A.G.; Data curation, K.G. and Y.V.; Formal analysis, K.G., Y.V., and A.G.; Funding acquisition, H.D., A.G.; Investigation, K.G.; Methodology, K.G., A.G.; Project administration, K.G., A.G.; Resources, K.G., A.G.; Software, K.G.; Supervision, H.D., A.G.; Validation, K.G., Y.V., K.P., H.D., and A.G.; Visualization, K.G.; Roles/Writing - original draft, K.G.; Writing - review & editing, K.G., Y.V., K.P., H.D., and A.G.

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identified: (1) vaccine promotion in targeted educational and outreach campaigns, (2) community-level support programs integrated with vaccine rollout, (3) rollout of a pan-genotypic, multivalent, or combination vaccine, and (4) cost-benefit analysis supporting the vaccine.

Conclusion: This forward-looking paper offers several recommendations to address potential gaps in HCV vaccination—from linkage with syringe exchange programs to economic analysis of vaccination program costs.

Keywords

Vaccination; Vaccination Campaign; Hepatitis C; People Who Inject Drugs; Vaccine Hesitancy

1. Introduction

Out of all infectious diseases in the United States, hepatitis C virus (HCV) has the largest mortality burden and was estimated to have impacted 1.0% of all United States adults in 2020 (approximately 2.5 million persons with HCV infections) [1], [2]. Historically, HCV was often transmitted by infected blood transfusions before changes to blood supply screenings in 1992 [3]. Most current HCV infections in the United States are linked with intravenous drug use in people who inject drugs (PWID), and case rates have continued to increase in the past decade until 2022 [1], [4], [5]. HCV case rates may be underestimated due to asymptomatic presentations, with a report from the Centers for Disease Control and Prevention (CDC) indicating that the true count of acute HCV infections may be several times larger than what is officially reported [6], [7].

HCV is now routinely cured with direct-acting antiviral agents (DAAs) [8]. These treatments are designed to target specific regions of key viral enzymes, curing infection in spite of HCV's profound genetic diversity. Although prices have dropped significantly since the 2010s, DAA treatments remain expensive, ranging from $10,000 to $40,000 (2018–2020 prices, prior to reimbursements from insurers) [9], [10], [11], [12]. Several disparities in HCV treatment with DAAs currently exist [13]. High-risk groups such as PWID are also less likely to receive insurance coverage for DAA treatments and face structural barriers when accessing treatment. HCV reinfection is also possible. Reinfection has been estimated to occur at a rate of up to 11.4 cases per 100 PWID each year [14]. This results in repeated DAA treatment cycles and additional cost to patients and insurers, with estimates ranging up to seven cycles of DAAs [15]. This is explained by T-cell exhaustion that occurs after infection, where pre-existing adaptive immunity is not sufficient to prevent reinfection (although DAA treatments partially mitigate this effect) [16], [17]. These treatment disparities, high treatment costs, and consequences of reinfection establish the need for a prophylactic HCV vaccine [2]. The creation of potential vaccine candidates for HCV has flourished in recent years, resulting in several preclinical vaccine candidates and one clinical trial [18], [19], [20]. Although no successful HCV vaccine has been created yet, momentum has been gained with the advent of mRNA technology that facilitated vaccine development for COVID-19 and respiratory syncytial virus (RSV). Additionally, HCV vaccine development may be accelerated with a controlled human infection model (CHIM) [21], [22], [23]. In the case of HCV vaccine development, CHIMs use the purposeful infection of human volunteers with vaccine candidates to demonstrate preliminary efficacy

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and then cure participants with DAAs [21]. In these trials, it is crucial to ensure that informed patient consent is properly obtained, especially from at-risk populations [24], [25]. Many PWID—especially younger individuals—display a notable willingness to participate in CHIMs or other randomized HCV vaccine efficacy trials [25]. This is of particular interest because the first HCV vaccine clinical trial was performed in uninfected PWID [19]. Due to T-cell exhaustion, the performance of a future vaccine may differ markedly between PWID who have been infected with HCV and PWID who have not been previously infected. This means that CHIMs that recruit PWID who have a wide range of immunological backgrounds have an important role in accelerating vaccine development. A prompt vaccine rollout may streamline the World Health Organization's goal of HCV elimination before 2030 [26].

However, even if a vaccine is created, barriers to vaccination could greatly limit uptake. For example, the COVID-19 pandemic revealed a climate of vaccine hesitancy, with particularly low rates of vaccine uptake and high expressed levels of COVID-19 and hepatitis B virus (HBV) vaccine hesitancy in PWID [27], [28], [29], [30], [31]. Existing work has focused on HCV vaccine development, but a systematic search in PubMed on January 1st, 2025—with keywords relating to barriers of vaccine rollouts and HCV—found no previous studies published since January 1st, 2015 that discuss potential challenges of a prophylactic HCV vaccine rollout (Supplement 1). Therefore, in this work, we aim to outline key barriers towards vaccine acceptance and make recommendations to overcome them. We advocate for the preparation of public and community health services in the present to facilitate a smooth and successful vaccination rollout with the future release of an approved HCV vaccine candidate.

With an unstructured integrative review, we synthesized literature from HBV and COVID-19 as historical precedents alongside currently existing barriers to HCV testing and treatment. These historical vaccination campaigns and current treatment interventions are used as proxies to assess challenges in future HCV vaccine uptake. Interventions that may impact HBV and COVID-19 vaccine hesitancy amongst PWID specifically have also been assessed with randomized controlled trials (Clinical Trials ID: NCT00244374) [32]. Many HBV-related interventions to increase vaccine acceptance and adherence to treatment in PWID may capture findings relevant to HCV vaccination [33]. Overall, we argue that lessons learned from the rollout of HBV vaccines, COVID-19 vaccines, and HCV testing and management initiatives may help public and community health professionals anticipate and overcome challenges in introducing HCV vaccination in vulnerable groups such as PWID, later expanding to the general population. These findings are noted alongside currently existing barriers to treatment.

2. Methods

In this work, we conducted an integrative narrative review to synthesize the literature related to barriers of vaccination, with a goal of anticipating challenges in HCV vaccine uptake [34]. PubMed and Google Scholar were used for searches. Findings from research in PWID in the United States were emphasized, but articles about challenges of vaccination in the general population and international articles were also considered. Searches emphasized COVID-19 and HBV vaccination campaigns and health behaviors and interventions related

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to HCV management. After this review, potential challenges that could decrease HCV vaccine acceptance in PWID were identified. These findings have been organized into thematic categories using precedents from COVID-19 and HBV vaccine rollouts as well as current interventions to address barriers in HCV management. Potential recommendations that may reduce the impact of barriers to HCV vaccination have been suggested. Perspectives from team members who work in HCV vaccine clinical trials, development, and outreach are also provided.

3.1 Results

In this section, we provide data about COVID-19 and HBV to contextualize our assessment of their vaccine rollouts in reference to HCV. We also assess potential challenges in HCV vaccination, denoting historical precedents from the COVID-19 vaccine rollout, HBV vaccine rollout, and HCV treatment access, as indicated by the literature. These challenges inform our recommendations alongside frameworks of health behavior.

3.2 Precedents to Potential Challenges

In Table 1 is a general comparison between COVID-19, HBV, and HCV. Each infection's duration is presented with the list price of potential treatments. Vaccine designs are considered. A non-exhaustive list of vulnerable populations is provided.

Challenges observed in COVID-19 and HBV vaccines along with HCV treatment are reported in Table 2. These challenges are organized into six conceptual domains: (1) structural and social barriers affecting PWID, (2) vaccine safety, efficacy, and relevance concerns, (3) multiple-dose attrition and vaccine fatigue, (4) media presentation and misinformation, (5) awareness and attitude towards infection, and (6) information framing and primary care linkage. Specific barriers to vaccination have been summarized within each category. We further elaborate on each of these barriers in 3.3.

3.3 Expected Challenges

Based on our review of the literature, we have identified the following key challenges for the rollout of an HCV vaccine and have grouped them into six categories below. Many specific barriers faced by PWID may play a role in reduced levels of vaccination, including key drivers of vaccine hesitancy [83]. We largely use findings from the literature on vaccine hesitancy in COVID-19 and HBV to provide evidence of these challenges and their potential roles in an HCV vaccine rollout, as appropriate. We also consider common barriers that prevent access to current HCV treatments, serving as barriers to HCV vaccination by extension [48].

3.3.1 - Challenge 1: Structural and social barriers affecting PWID

Many disparities in HCV treatment with DAAs have persisted through the late 2010s and early 2020s. Location-based disparities of DAA treatments for HCV indicate a need for greater coverage in rural and non-urban communities [37], [38]. HCV also affects more racial and ethnic minorities in the United States, and disparities in chronic HCV management and treatment exist along these lines [41], [42]. Finally, age- and sex-related

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disparities may exist among PWID for DAA access to HCV management, resulting in less coverage in women and younger adults [41]. Similar health disparities may impact HCV vaccination accessibility and delivery.

Stigma and discrimination also contribute to medical mistrust that is prevalent amongst vulnerable groups such as PWID [43], [84], [85], [86]. An overarching distrust of vaccination is higher in PWID than in the general population [30]. High levels of distrust have been shown to correlate with reduced adherence to DAA treatments in PWID with HCV [45]. Structural barriers in the medical system also exist. For instance, policies that require sobriety or limited injection drug use for treatment restrict access to HCV treatment in PWID [49]. These barriers are commonly cited by PWID, alongside stigma, discrimination, personal embarrassment, and anxiety, fear, or dislike of the medical establishment also serving as reasons that PWID currently forgo HCV treatment [48]. Negative interactions with healthcare systems may affect the choice to undergo treatment or receive vaccination later in life [85], [86]. Additionally, many PWID with HCV encounter competing priorities that limit access to HCV treatment and are correlated with increases in high-risk behaviors [87]. Among PWID, barriers such as food and housing insecurity are prevalent, and these needs often take precedence over HCV treatment [87], [88], [89], [90]. These barriers may cause PWID to experience logistical challenges in seeking treatment, such as lack of extra time to attend appointments or breakpoints in accessing care. When priorities compete, vaccine availability and other social determinants of vaccine outcomes (e.g., reliable transportation) may serve as barriers towards vaccine acceptance.

Other logistical and structural barriers may also influence the effectiveness of a universal HCV vaccine rollout. One concern is the efficacy of current vaccination programs aimed at PWID, specifically at locations where PWID seek care, such as syringe service programs (SSPs). Findings from interviews and surveys of SSPs in COVID-19-specific and general vaccination programs highlight resource limitations, logistical issues, and concerns from PWID about vaccination. Resource limitations include lack of licensed staff, vaccine supplies, storage equipment, and tracking systems for multiple-dose vaccines [50], [51]. Furthermore, current geographic disparities in DAA treatments may reflect future trends in vaccine availability, alienating PWID from vaccination rollout efforts. Across geographic regions, insufficient community engagement and partnership with high-risk groups of PWID could further restrict vaccine rollout. Many determinants of HCV treatment may also affect vaccination, including transportation challenges, outreach difficulties, and other linguistic or communication barriers [47]. HCV vaccine rollouts must consider the impact of ongoing structural barriers to prevention and treatment in vulnerable populations.

3.3.2 - Challenge 2: Vaccine safety, efficacy, and relevance concerns: The advent of mRNA technology to create COVID-19 vaccines revealed a unique form of vaccine hesitancy involving fear of novel technologies, despite the low risk of side effects [52]. Novel technology bias may increase hesitancy towards a potential vaccination for HCV. Additionally, the possibility of adverse effects can decrease vaccine acceptance and increase vaccine fear [91]. Knowledge of rare side effects has been shown to mediate HBV vaccine intentions in Chinese adults over the age of 24 [92], which is notable given that China has the largest global HBV burden [93]. Additionally, specific vaccine ingredients

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used as adjuvants may also contribute to hesitancy, even among populations like healthcare workers with high vaccine knowledge, as in the case of the HBV vaccine rollout [94]. It is possible for any type of HCV vaccine candidate to result in rare side effects or use particular adjuvants that could create similar misperceptions about vaccine safety or efficacy. Finally, lack of knowledge about HCV prevention, general vaccination fear, or fear of testing positive for HCV could prevent PWID and other high-risk individuals from seeking care, treatment, or vaccination.

The acceptance of HCV vaccination may also be impacted by unique safety and efficacy concerns when attempting to broaden to the general population, particularly among caretakers of youth and adolescents. For example, human papillomavirus (HPV) infection is often associated with sexual activity. Parents may choose not to vaccinate their children for HPV because they believe the vaccine could increase the likelihood of youth sexual activity, regardless of evidence to the contrary [95], [96]. Cultural beliefs, religious beliefs, and religiosity may play a role in mediating these effects [97], [98]. Similarly, HCV is a bloodborne infection primarily transmitted by contaminated injection equipment. Parents may opt out of prophylactic vaccination out of a desire to prevent their children from injecting drugs or due to the belief that HCV vaccination would be unnecessary or irrelevant.

This coincides with caretaker concerns about the safety of an HCV vaccine in the general population. For example, at birth, HBV immunizations are given to prevent HBV transfer to infants. Infant non-vaccination may begin as early as an at-birth dose of HBV immunization. Although refusal of HBV vaccines at birth has substantially decreased over the past decade, vaccine hesitancy may persist at an average estimated rate of 3.5% (Washington DC) [99]. Later in childhood, parents often justify HBV vaccine refusal by referencing safety concerns or misinformation found on the internet [58]. Infants who are not vaccinated for HBV are more likely to miss other scheduled vaccinations, suggesting the persistence of negative sentiments about vaccines [59]. Data-driven, culturally relevant educational sessions for parents may help contribute to reduced HBV vaccine hesitancy and refusal [100]. While vertical transmission of HCV through birth is possible, it is incredibly rare, and an HCV vaccine created for the general population may or may not be administered in infancy or childhood [101], [102]. In spite of these differences, analogous concerns about the safety, efficacy, and relevance of HCV vaccines rolled out to the general population may pose a significant challenge for vaccine uptake.

3.3.3 - Challenge 3: Multiple dose attrition and vaccine fatigue: A vaccine for HCV could require multiple doses, as many other vaccines do, causing reduced acceptance and/or completion. Multiple-dose hesitancy or attrition is a demonstrated challenge in HBV vaccine uptake; evidence suggests that compared with a baseline speed, accelerated HBV vaccine schedules across multiple doses can cause increased vaccine acceptance and uptake [61]. Another issue with a potential HCV vaccine is the possibility of delivering boosters due to waning immunity [62]. While booster vaccines for HBV are often unnecessary, in the case of COVID-19, where boosting is commonplace, booster hesitancy disproportionately affects PWID [63]. PWID may be less likely to engage in HCV vaccination uptake across multiple, spaced-out doses or boosters.

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Additionally, it is possible that PWID who have received other vaccines to completion, such as the HBV and COVID-19 series, experience vaccination fatigue. Vaccination fatigue occurs when receiving a vaccine decreases an individual's willingness to engage in future vaccination efforts due to burnout, lack of convenience, or apathy [64]. This pattern of non-adherence can also be found to currently impact HCV testing and treatment rates, such as absence of key appointments or missing lab work [47]. These factors could present a barrier to achieving sufficient HCV vaccination coverage rates among PWID.

3.3.4 - Challenge 4: Media presentation and misinformation: Vaccine sentiments may be affected by presentation in the mass media or social media. Negative vaccine presentation on social media has been shown to play a role in explaining vaccination disparities by geographic area [103]. After the COVID-19 pandemic, researchers observed widespread increases in negative vaccine sentiment, vaccine disinformation, and support of conspiracy theories [104], [105]. These effects can dampen vaccine acceptance and could promote general hesitancy towards an HCV vaccine. Even before the COVID-19 pandemic, HBV vaccines have been particularly noted for their poor reception by the media in the United States compared to other vaccines; negative media coverage has been attributed to specific adjuvants used as ingredients to procure the HBV vaccine [54]. While current HCV vaccine candidates use a range of adjuvants, similar increases in negative sentiment may follow with the creation of an HCV vaccine.

Additionally, vaccine misinformation has also been found on social media. Exposure to misinformation has been associated with vaccine hesitancy during the COVID-19 pandemic [65]. This information has often been found to conflict with information from science-based news sources in content or mimic the presentation of media information, which may create a sense of confusion for individuals considering vaccination [66]. Conspiracy theories about vaccines have also proliferated with the rise of the internet and are linked with vaccine hesitancy worldwide [66], [106]. Even before the COVID-19 vaccine rollout, various vaccine-related conspiracy theories have proliferated (e.g., the false causal link between childhood vaccination and autism) [107]. Many conspiracy theories have emerged since the COVID-19 pandemic, and evidence indicates that a significant proportion of PWID display vaccine hesitancy related to this type of vaccine disinformation [30], [108]. These challenges may play a role in correctly disseminating information about an HCV vaccine.

3.3.5 - Challenge 5: Awareness and attitude towards infection: If an HCV vaccine were to be made available, lack of HCV-related knowledge and one's personal infection status could serve as barriers to a risk-based vaccine rollout in certain groups [109], [110]. HCV has a long chronic phase with few symptoms, leading infected individuals to underestimate its risk and need for prevention. As of 2023, an estimated 840,000 people in the United States were not aware of their chronic HCV infection, with disproportionately low awareness amongst foreign-born residents from countries in Asia or Latin America and individuals with a low socioeconomic status [109], [110]. Some of these trends are also reflected in studies of HBV infection awareness and may coincide with disparities in HBV vaccination [40].

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This challenge may differ in PWID. Some PWID or other vulnerable individuals may perceive that contracting HCV is unavoidable due to their lifestyle and high-risk environment. HCV fatalism—the perceptions that (1) HCV infection is inevitable with continued injection drug use and (2) preventive interventions or curative treatments are not worth receiving, regardless of vaccine or treatment efficacy—is noted as a key barrier to existing treatment efforts in PWID [48], [75], [76]. Evidence also indicates that fatalistic attitudes predict general apathy towards vaccination, vaccine hesitancy, or lack of infection screening [46], [72], [74], [111], [112]. Intervention programs that specifically address an individual's health locus of control may reduce the impact of this challenge on HCV vaccination. For instance, interventions that emphasize the individual responsibility to reduce the spread of HCV to others may change behaviors among PWID [28], [113]. These trends in current HCV management may persist with the rollout of a vaccine.

HCV vaccine candidates may also be non-sterilizing of HCV, contributing to potential hesitancy. Vaccine candidates in development are typically designed to elicit a strong T-cell response. Successful vaccines effectively promote viral clearance during the chronic phase of the infection, but this is not necessarily associated with the prevention of HCV infection. This could uniquely influence attitudes about perceived utility amongst the general population, PWID without HCV, and PWID who have a history of HCV infection. The general population and PWID with no known history of HCV infection may have low awareness and knowledge about HCV in general, and this may serve as a primary driver of vaccine hesitancy. Conversely, many PWID who have had their HCV treated with DAAs may be aware of their high reinfection risk, which could reinforce fatalistic attitudes. PWID often exhibit partial T-cell exhaustion even following DAA treatment courses, with adaptive immune responses failing to protect them from reinfection [17]. In these individuals, it is possible that a vaccine candidate also fails to stimulate exhausted T cells or elicit a protective immune response after vaccination or only results in partial benefits [19]. This may lead to HCV fatalism and the perception that a vaccine is not worth receiving.

3.3.6 - Challenge 6: Information framing and primary care linkage: Information framing from trusted sources is shown to be important in vaccine uptake. We use the example of the HPV vaccination campaign. While HPV may cause many types of cancer (e.g., cervical cancer), many individuals who are vaccine hesitant are not aware of this when considering vaccination [114], [115]. Mass media coverage and framing effects may impact vaccination attitudes, and the characterization of HPV vaccines as cancer-preventing technologies has informed vaccine intentions [116].

However, we note that in hepatitis A virus (HAV) and HBV vaccination, evidence regarding the impact of framing effects on vaccine intent and uptake is mixed, but a healthcare professional's strong, direct recommendation to vaccination can increase vaccine intent [79], [80]. Clinician recommendation—particularly in primary care—can be an important protective factor of vaccine acceptance that can counteract concerns and challenges, especially in PWID [53]. For individuals with HCV, recommendation of HAV and HBV vaccination by primary care professionals has been suggested to improve vaccine uptake [117]. This need directly translates to HCV vaccine uptake.

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Linkage with primary care is also a challenge. State-funded efforts to implement educational campaigns to raise awareness about HCV in high-risk groups such as PWID may have an insubstantial impact on primary care clinicians' behaviors in practice [82]. Healthcare professionals who recommend HBV and HCV vaccines would have to consider multiple high-risk audiences, including both PWID and other healthcare workers [118], [119], [120].

Importantly, similar trends are observed in the HCV screening landscape. The CDC's Advisory Committee on Immunization Practices (ACIP) currently recommends testing after possible HCV exposures in the general population, but a lack of targeted messaging and direct clinician recommendation has been associated with reduced screening and could similarly impair HCV vaccine uptake [121].

4.1 Recommendations

Overall, we found several heterogeneous barriers to HCV vaccine acceptance that exist in the general population as well as in PWID. We propose the following theory-informed recommendations:

4.1.1 - Recommendation 1: Vaccine promotion in targeted educational and outreach campaigns

We recommend public and community health organizations to implement targeted educational campaigns for both clinicians and high-risk groups, such as PWID. These programs could aim to increase vaccine knowledge, trust in vaccine safety, acceptance of a potential HCV vaccine, or overall health locus of control [122]. To equitably prevent the greatest number of infections, risk-based approaches are often used when designing education, outreach, or intervention programs—as well as vaccine rollouts [123]. Campaigns organized by public and community health professionals may utilize clinician recommendations and focus on presenting vaccination options informed by research on health behaviors in vulnerable groups. This promotion may occur alongside the current CDC recommendations to address structural barriers to vaccination by providing culturally and linguistically sensitive, community-oriented education campaigns for testing and curative treatment—and programs designed with multiple types of community and clinical perspectives [47]. Vaccine promotion may be added to these materials, and in audiences of PWID, public health authorities could frame vaccination to prevent fatalistic attitudes toward infection and improve compliance with treatment. This can use frameworks of health behavior. For example, HCV causes cancer (hepatocellular carcinomas), and the vaccine may be presented as a form of cancer prevention according to frameworks such as the Health Belief Model (HBM) or Protection Motivation Theory [124], [125]. This may also improve the overall uptake of a prophylactic vaccine administered to the general population after high-risk communities such as PWID. We stress the importance of community and primary care physician engagement in the process of this public-health-driven outreach. More outreach, training, and education rooted in social and behavioral theory (e.g., HBM, 5C Model) may also help healthcare professionals effectively promote HCV vaccination before the vaccine's release [126].

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4.1.2 - Recommendation 2: Community-level support programs integrated with vaccine rollout

We recommend an incremental, risk-based rollout of an HCV vaccine led by public health authorities, initially targeting high-risk groups such as PWID and healthcare workers [127]. Subsequently, the vaccine rollout could gradually expand to the general population. In the risk-based phase of HCV vaccination, public and community health professionals could integrate community-level support programs (e.g., field clinics, syringe exchange programs, routine testing, or connection with curative treatment options) with vaccine rollouts. An ideal risk-based phase of the rollout might also consider immunological heterogeneity amongst PWID. For instance, PWID who experience HCV reinfection could be prone to T-cell exhaustion [128], and tailored follow-up appointments or community-based programs may benefit these individuals [14]. In the context of supporting and improving currently existing vaccination programs that target PWID, there are multiple ways to address resource limitations, including increasing funding to SSPs and increasing collaboration between SSPs, local and state health departments, and licensed vaccination providers to help address staffing and equipment needs. Vaccine hesitancy can be addressed through fostering a trusting relationship between SSP staff and PWID. We highlight the importance of community partners in this work who are familiar with structural barriers to access and compliance, safety concerns, and vaccine misinformation as well as the different immunological challenges that individual patients face. These initiatives may be driven by theoretical frameworks such as the HBM. Current CDC recommendations stress collaborations where HCV prevention services are integrated between community members (especially trusted messengers in the community) and academic and healthcare professionals. Vaccination rollout infrastructure may be integrated into these pipelines.

4.1.3 - Recommendation 3: Rollout of a multivalent or combination vaccine

We recommend that vaccine developers conduct research into the possibility of creating a pan-genotypic, multivalent, or combination vaccine. Of particular note is the possibility of a pan-genotypic vaccine. With several primary genotypes and dozens of subtypes, HCV exhibits a wide degree of genetic diversity, presenting a challenge for vaccine development [129]. Methods of pan-genotypic vaccine development take advantage of unique immunological strategies (e.g., targeting conserved epitopes) [130], [131]. Pan-genotypic vaccine design has the potential to standardize the vaccine rollout across different populations of PWID. A vaccine candidate with broad coverage could improve vaccine acceptance by alleviating concerns related to vaccine efficacy. Additionally, combination vaccines may be of interest. This could protect against HCV alongside viruses such as HAV and HBV, similar to combinations like the Diphtheria Tetanus Pertussis (DTaP) vaccine. Prophylactic HAV and HBV childhood vaccines exist, and combination vaccines for HAV and HBV demonstrate remarkable efficacy, safety, and duration of immunity (i.e., Twinrix, manufactured by GlaxoSmithKline Biologicals) [132], [133]. However, while HAV and HBV vaccination rates are increasing, they remain low (at around 40% or less in the general population), and they generally decrease as a cohort’s age increases [134]. Superinfection between hepatitis viruses (such as HCV and HBV) may increase the downstream risk of cirrhosis or hepatocellular carcinomas [135]. Combination vaccines could increase the uptake of HAV and HBV vaccines in PWID with HCV—where uptake has been estimated

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at less than 10%—and mitigate long-term health risks [136], [137]. Vaccinologists' research into the feasibility and/or development of multivalent vaccines may result in the creation of a combination vaccine, promoting HCV vaccine uptake.

4.1.4 - Recommendation 4: Cost-benefit analysis supporting the vaccine

We recommend that vaccine developers and researchers build a case for the novel vaccine, addressing different audiences: clinicians, healthcare organizations, public health authorities, and advisory committees. Particularly important is the CDC's Advisory Committee on Immunization Practices (ACIP) that could ensure it is covered by healthcare payers. ACIP recommendations have previously been used to guide decisions on universal HBV and COVID-19 vaccine rollouts [138], [139], [140]. Different stakeholders may focus on different outcome metrics, such as unit cost, cost per quality-adjusted life year, and overall program fiscal return. This could inform decision-making and allocation of limited financial resources for a hypothetical HCV vaccine rollout. Researchers could support the analysis exploring the population-level effects of the vaccine. Models could use predictors specific to HCV transmission and PWID, such as syringe-sharing behaviors, risk of reinfection and whether patients are receiving opiate substitution therapy [141], [142], [143].

4.2 Discussion

This review has summarized several barriers to potential HCV vaccine uptake in PWID, expanding to the general population as necessary. Similar challenges—such as stigma, structural barriers, and mistrust in health systems—have impacted COVID-19 and HBV vaccination rollouts as well as HCV management initiatives. These challenges are likely to repeat in the event of an HCV vaccine rollout. However, HCV's curability, association with PWID, and expense of current treatment options introduce further complexity to our predictions. Models of health behavior may be used to supplement existing frameworks of vaccine hesitancy in the case of an anticipated HCV vaccine rollout. An HCV vaccine rollout may benefit from targeted outreach campaigns, integration with community-based testing and treatment initiatives, multivalent vaccine creation, and promotion through federally funded programs. These recommendations emphasize the importance of modifying vaccine rollout strategies to reflect population-specific challenges faced by distinct groups of PWID—individuals who differ both behaviorally and immunologically. This ranges from designing messaging to address HCV fatalism to developing guidelines that consider reinfection risk and various levels of natural immunity.

4.3 Limitations

This is a forward-looking paper written before the rollout of HCV vaccines. Challenges and proposed solutions outlined in this integrative review represent the current state of the literature and may not address barriers to HCV vaccination at the time of the vaccine's creation. This paper was written in a narrative review style—we primarily use the COVID-19 and HBV vaccine rollouts as well as HCV treatment interventions as examples and recognize that unforeseeable barriers may occur as an HCV vaccine candidate is rolled out. We often discuss the behavioral and social factors that may influence vaccine hesitancy and want to emphasize the speculative nature of our claims. Various associations are found

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between such factors and vaccine hesitancy throughout the literature, and we note that many causal explanations of vaccine hesitancy cannot be directly assumed. Because of this, we stress that more data in the field is needed and that our summary of challenges and recommendations is non-exhaustive, as a limited number of studies on HCV vaccination have been conducted. Notably, the HCV vaccine development process itself mediates many of the downstream challenges discussed in this paper. For instance, whether or not a vaccine candidate maintains the same efficacy in individuals who have been treated for HCV may influence disparities in vaccine hesitancy between different groups. Because of this, robust immunological, genotypic, and sociobehavioral data of PWID who have HCV at multiple geographic levels would provide more insight into what challenges should be prioritized. As recommended by the CDC, recognition of high-risk groups at the local, state, and federal levels is important to inform current preventive strategies, and it is also important when considering vaccination [144]. Our review does not address these data sources directly, and as such, we have not prioritized specific challenges.

5. Conclusion

Overall, we outline key challenges and recommendations that could play a role in HCV vaccine rollouts as a vaccine becomes available. The overall success of an HCV vaccine campaign depends on anticipating key barriers that may impact vaccine uptake. This is accomplished in this paper by analyzing historical trends in HBV and COVID-19 vaccine hesitancy as well as current barriers in HCV treatment that extend to vaccine hesitancy. We advocate for multidisciplinary research that integrates findings from vaccine development with downstream policy and rollout efforts.

In this manner, our findings highlight that an HCV vaccine rollout must be carefully designed in order to meet the needs of multiple sub-populations of PWID as well as the general population. Many factors restrict the generalizability of a vaccine rollout—from immunological variability related to previous HCV exposure to the wide range of competing priorities often faced by PWID, such as food or housing insecurity. To address this, we recommend the creation of outreach campaigns targeted towards unique sub-populations, integration with community-level support programs that facilitate vaccine rollouts, the development of pan-genotypic vaccines that may mitigate challenges with HCV’s genetic diversity, and cost-benefit analyses to support the allocation of vaccine rollout resources. In tandem, these potential solutions may help address each challenge related to HCV vaccine uptake and prepare for the eventual release of a vaccine candidate.

Supplementary Material

Refer to Web version on PubMed Central for supplementary material.

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Table 1.

Vaccination landscape overview of COVID-19, HBV, and HCV

Aspect	COVID-19	HBV	HCV
Existence of vaccine	Yes	Yes	No
Peak vaccination rate estimates of all U.S. adults	~70%	~30% [35]	N/A
Type of vaccine or vaccine candidates	mRNA, viral vector	Recombinant protein	DNA, protein subunit, viral vector[a]
Infection duration	Acute; can result in long-term complications	Acute or chronic	Acute or chronic
Estimated list price per course of treatment duration (before insurance coverage)	$1,000–$2,000	$20–$200, $1,000–$5,000 (annual total cost)[b]	Up to $40,000 (DAA treatments)[c]
Populations of high prevalence in the U.S.	Less vaccinated communities, healthcare and other frontline workers, racial and ethnic minorities	Foreign-born individuals, individuals with multiple sex partners, men who have sex with men (MSM), healthcare workers, PWID	Individuals with multiple sex partners[d], MSM[d], healthcare workers, PWID

[a]Denotes HCV vaccine types in early clinical phases; vaccine designs in preclinical phases include DNA, mRNA, protein subunit, viral-like particle, and peptide vaccines

[b]Antiviral treatment is maintained for life in most patients chronically infected with HBV

[c]Curable with one course of DAA treatment in most cases

[d]Uncommon in comparison to HBV

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Table 2.

Challenges observed in COVID-19 and HBV vaccines and HCV treatments, grouped into conceptual domains

Challenge	Challenge Precedent
Structural and social barriers affecting PWID	
Geographic-level disparities	COVID-19[a] [36], HCV treatment[b] [37], [38]
Demographic-level disparities	COVID-19 [39], HBV[c] [40], HCV treatment [41], [42]
Stigma and discrimination	HCV treatment [43], [44]
Distrust of public health institutions	COVID-19 [28], HCV [45]
Competing barriers (e.g., food insecurity)	HBV [46], HCV treatment [46], [47], [48]
Policy barriers prior to intervention	HCV treatment [48], [49]
Logistical barriers at access points	HCV treatment [47], [48], [50], [51]
Vaccine safety, efficacy, and relevance concerns	
Novelty bias against vaccines	COVID-19 [52]
Concerns about vaccine side effects	COVID-19 [28], [53]
Concerns about vaccine ingredients	HBV [54], [55]
Concerns in caretakers of youth	COVID-19 [56], [57], HBV [58], [59]
Multiple-dose attrition and vaccine fatigue	
Multiple-dose attrition	COVID-19 [60], HBV [61]
Booster hesitancy	COVID-19 [62], [63]
Vaccine fatigue	COVID-19 [64]
Media presentation and misinformation	
Negative mass media coverage	HBV [54]
Social media and misinformation	COVID-19 [65], [66]
Conspiracy theories	COVID-19 [30], [67]
Awareness and attitude towards infection	
Awareness of infection status	COVID-19 [68], HBV [69], HCV treatment [70], [71]
Fatalistic attitude towards cure or prevention	COVID-19 [72], [73], [74], HCV treatment [75], [76]
Information framing and primary care linkage	
Framing of health intervention	COVID-19 [77], [78], HBV [79], [80], HCV treatment [81]
Lack of integration or recommendation amongst primary care professionals	HBV [79], HCV treatment [47], [82]

[a]Challenge found in the COVID-19 vaccine rollout

[b]Challenge found in HCV intervention

[c]Challenge found in HBV vaccine rollout or intervention